

Total No. of Questions : 6]

P613

SEAT No. :

[Total No. of Pages : 2

BE/Insem/APR - 246
B.E. (Computer Engineering)
INFORMATION AND CYBER SECURITY
(2015 Pattern) (Semester - II)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Solve Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Explain Operational Security Model for Network Security. **[5]**

b) How Information Security attacks are classified? Give example for each. **[5]**

OR

Q2) a) What are different security policies? Explain. **[5]**

b) List the differences between Security & Privacy. **[5]**

Q3) a) Use PlayFair Cipher to encrypt the message "This is a columnar transposition". Use key - APPLE. **[5]**

b) Explain following algorithm modes **[5]**

i) ECB

ii) OFB

OR

Q4) a) Explain the operation of DES algorithm in detail. **[5]**

b) Explain Monoalphabetic & Polyalphabetic ciphers with appropriate examples. **[5]**

P.T.O.

Q5) a) Find the key exchanged between Alok and Bobby considering following data.

i) $n = 11$

ii) $g = 5$

iii) $X = 2, Y = 3$

Find value of A, B and secret key K.

[5]

b) What is Kerberos? Explain operation in detail.

[5]

OR

Q6) a) Given two Prime Numbers $P = 17$ & $Q = 29$ find out N, E, & D in an RSA encryption process.

[5]

b) Explain in details the need & implementation of one way hash function (MD5).

[5]

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